

What is CSS Box Model?

-> The CSS Box Model is a concept used in CSS to describe how every HTML element is displayed as a rectangular box.

-> Each element on a webpage is treated like a box that contains:

1. Content
2. Padding
3. Border
4. Margin

-> The box model helps us control:

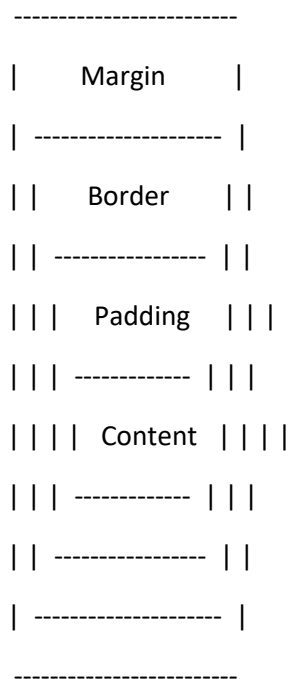
Spacing

Size

Layout

Alignment of elements

Structure of CSS Box Model:



4 Main Components of CSS Box Model:

-> There are 4 main components of box model are:

1. Content
2. Padding
3. Border
4. Margin

1. Content:

-> The content area is the actual part where text, images, videos, or other elements appear.

2. Padding:

-> The space between content and border is called as padding.

OR

-> The surrounding area of the content is called as padding.

-> It creates inner spacing inside the element.

-> Syntax: padding: value;

-> Individual Padding:

padding-top: 10px;

padding-right: 20px;

padding-bottom: 10px;

padding-left: 20px;

-> padding is the shorthand property for the all 4 sides.

1. padding: 10px; // All side - TRBL (Top, Right, Bottom, Left)

2. padding: 10px 20px;

First value - (Top Bottom), Second value - (Left Right)

3. padding: 10px 20px 30px;

First value - Top

Middle value - Left and Right

Third value - Bottom

4. padding: 10px 20px 30px 40px;

First value - Top

Second value - Right

Third value - Bottom

Fourth value - Left

3. Border:

-> Border is the line surrounding padding and content.

-> Syntax: border: border-width border-style border-color;

-> Ex: border: 2px solid black;

-> Different Border Styles:

1.solid

2.dashed

3.dotted

4.double

5.groove

6.ridge

7.inset

8.outset

9.none

4. Margin:

-> Margin is the space outside the border.

-> It creates space between elements.

-> Syntax: margin: value;

-> Individual Margin:

1.margin-top

2.margin-right

3.margin-bottom

4.margin-left

-> margin is the shorthand property for the all 4 sides.

1. margin: 10px; // All side - TRBL (Top, Right, Bottom, Left)

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Second value - Right

Third value - Bottom

Fourth value - Left

What is margin: auto?

-> margin: auto is commonly used to center a block element horizontally.

-> Conditions for margin:auto

-> To work properly:

1. Element must be a block element

2. Width should be given

What is box-sizing Property?

-> The box-sizing property defines how the total width and height of an element are calculated.

-> We can control the size of the box.

-> Syntax: box-sizing: content-box (default value) | padding-box | border-box;

-> Example: 1. box-sizing: content-box (default value)

Total Width = width + padding + border

-> Example: 2. box-sizing: border-box

Total Width = content + padding + border

Why Developers Use border-box:

-> Most developers use:

```
* {  
  margin: 0;  
  padding: 0;  
  box-sizing: border-box;  
}
```

-> Because it makes sizing easier and avoids layout issues.

-> It is also called as Reset Code.

What is border-radius Property?:

-> The border-radius property is used to create rounded corners.

-> Syntax: border-radius: value;

-> Circle Example: border-radius: 50% or 100%;

-> Individual Sides:

1.border-top-left-radius

2.border-top-right-radius

3.border-bottom-right-radius

4.border-bottom-left-radius

-> Shorthand:

border-radius: 10px 20px 30px 40px;

Order: Top-left Top-right Bottom-right Bottom-left